Liangyu WU | Curriculum Vitae

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Education Background

Stanford University — **California, U.S.**

Ph.D. Student in Physics (Advisor: **Dr. Julia Gonski** & **Dr. Dong Su**) — Sep. 2024-Present

- Research area: Experimental Particle Physics

Shanghai Jiao Tong University — **Shanghai, China**

Bachelor of Science in Physics, School of Physics and Astronomy — Sep. 2020- Jun. 2024

- Overall GPA: 86.1/100 (A-); Physics-related GPA: 90.1/100 (A)
- Coursework: *Quantum Mechanics*, *Electrodynamics*, *Equations of Mathematical Physics*, *Professional Experiment*, *Introduction to Nuclear and Particle Physics*, *Methods of Experimental Nuclear and Particle Physics* (All graded A's)

University of Maryland, College Park — **Maryland, U.S.**

Visiting International Student — Aug. 2023-Dec. 2023

- Overall GPA: 3.925/4.0

Research Experience

SLAC National Accelerator Laboratory — **California, U.S.**

Particle Physics and Astrophysics — Jan. 2025-Present

Research Assistant (Supervisor: **Dr. Julia Gonski** & **Dr. Ariel Schwartzman** & **Dr. Dong Su**)

– **Research on the ATLAS Experiment at CERN**

- Foundation model development with ATLAS Run 2 proton-proton collision data.
- Advancing LAr timing for primary vertex t_0 reconstruction towards 5D calorimetry with high precision timing.
- GigaBit Cable Receiver (GBCR) ASIC QC&SEU testing for the HL-LHC ITk upgrade.

– **Research on the Future Colliders**

- On-chip machine learning in future advanced detectors (dual-readout calorimeters, drift chambers, etc.).
- R&D for embedded FPGA (eFPGA) technology in advanced subsystems for future colliders.
- Exploration of R&D on dSiPM technology for future detectors.

SLAC National Accelerator Laboratory — **California, U.S.**

Particle Physics and Astrophysics — Sep. 2024-Dec. 2024

Research Assistant (Supervisor: **Prof. Spencer Gessner**)

– **Research on the FACET-II**

- Enhancements to the X-Band TCAV GUI and associated analysis tools.

University of Maryland, College Park — **Maryland, U.S.**

Department of Physics — Aug. 2023-Aug. 2024

Undergraduate Research Student (Supervisor: **Prof. Christopher Palmer** & **Prof. Sarah Eno**)

– **Research on the Dual-Readout Calorimetry**

- Investigation of single-particle responses through Geant4 simulations across diverse calorimeters.
- Formula derivation which predicts the dual-readout-corrected energy from scintillator and Cherenkov signals alone.

Shanghai Jiao Tong University — **Shanghai, China**

School of Physics and Astronomy — Oct. 2021-Feb. 2024

Undergraduate Research Assistant (Supervisor: **Prof. Yue Meng**)

– **R&D for a Novel Radon Detector**

- Designed and constructed the Radic detector and determined the diffusion coefficient of various membrane materials.

PandaX Collaboration **Shanghai & Sichuan, China**
Undergraduate Research Student Oct. 2021-Feb. 2023

– Work in Ultra-low background technique R&D group
- Measured radon emanation rates across diverse materials for the PandaX and JUNO experiments.
- Conducted ultra-low radioactive surface treatments of materials for TPC assembly in PandaX-4T experiment.

– Work in PMT group
- Performed PMT high-voltage testing to ensure standard operating currents and assessed parameters.
- Participated in the installation of veto PMTs and high-voltage cables for TPC-bottom PMTs.

Tsung-Dao Lee Institute **Shanghai, China**
Astronomy and Astrophysics Division Mar. 2023-Aug. 2023
Undergraduate Research Student (Supervisor: **Prof. Masahiro Ogihara**)

– Research on the stability of unevenly spaced planetary systems
- Conducted N-body simulations to study the planetary system stability.
- Demonstrated that using evenly spaced models can overestimate orbital stability time in natural systems.

Publications

[1] D. Yilmaz, **L. Wu**, J. Gonski, Edge Machine Learning for Cluster Counting in Next-Generation Drift Chambers, **[Accepted by NeurIPS 2025]**
[2] S. Eno, **L. Wu** *et al.*, On the resolution of dual readout calorimeters, **NIM A,Volume 1083, 2026, 171080, ISSN 0168-9002.**
[3] **L. Wu**, L. Si *et al.*, Design and Experimental Application of a Radon Diffusion Chamber for Determining Diffusion Coefficients in Membrane Materials, **JINST, 20, no.03, P03031 (2025)**
[4] S.V. Chekanov, S. Eno, S. Magill, C. Palmer, **L. Wu**, Geant4 simulations of sampling and homogeneous hadronic calorimeters with dual readout for future colliders, **NIM A,Volume 1072, 2025, 170200, ISSN 0168-9002.**
[5] S. Yang, **L. Wu** *et al.*, The stability of unevenly spaced planetary systems, **ICARUS, Volume 406, December 2023, 115757**

Conference Talks

1. [Oct 2025] **Coordinating Panel on Advanced Detectors (CPAD) 2025**
   - *Developing Digital Silicon Photomultiplier (dSiPM) Specifications for a High-Granularity Calorimeter with Simulations*
   - *Embedded ML Solutions for Real-time Processing in Future Drift Chambers*
2. [Jul 2025] **US ATLAS Summer Workshop 2025**
   - *Enhancing Timing Information Reconstruction with LAr Calorimeter*
3. [May 2025] **ITk Week 2025**
   - *Recent Results of GBCR3 Testing at SLAC*

Skills

Technical: C++, Python, CERN ROOT, Geant4, SolidWorks, COMSOL, LaTeX
Hardware: Ultra-low radioactive technology, High-voltage technology, PCB debugging
Language: English (Fluent), Mandarin (Native)

Extra-Curricular Experience

Student Union of School of Physics and Astronomy **Shanghai, China**
Co-President of the Student Union Dec. 2021-Dec. 2022

– Spearheaded the execution of events and promotions for the Student Union.

Selected Honors and Awards

2023 Best Report First Prize, TDLI Astro-Division Winter Camp (5 out of 96 participants)
2022 NetClass Education and Research Fund Scholarship: First Prize, Shanghai Jiao Tong University (Top 5%)
2021 and 2022 Academic Excellent Scholarship, Shanghai Jiao Tong University (Top 10%, twice)
2021 Chun Geng Education Scholarship, Shanghai Jiao Tong University, School of Physics and Astronomy (Top 5%)